THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $4x^2 + 12xy + 9y^2$	$(a+b)^2 = a^2 + 2ab + b^2$
2	**(a)** $9x^2 - 12xy + 4y^2$	$(a-b)^2 = a^2 - 2ab + b^2$
3	**(d)** $x^2 + 4x - 21$	$(x+a)(x+b) = x^2 + (a+b)x + ab$
4	**(b)** 10609	Write $103 = 100 + 3$
5	**(b)** $(x+4)$	$a^2 + 2ab + b^2 = (a+b)^2$
6	**(c)** $(x+2)(x+5)$	Find two numbers with the given sum and product.
7	**(b)** 53	$x^2 + y^2 = (x+y)^2 - 2xy$.
8	**(d)** $a^3 + 3a^2b + 3ab^2 + b^3$	Standard cube identity.
9	**(c)** $9x^2 + 24xy + 16y^2$	$(a+b)^2 = a^2 + 2ab + b^2$
10	**(a)** $4x^2 - 20xy + 25y^2$	$(a-b)^2 = a^2 - 2ab + b^2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Use the square identity (1); $= 9x^2 + 30xy + 25y^2$ (1).	1+1
12	$104 = 100 + 4$ (1); $104^2 = 10816$ (1).	1+1
13	Use $(x+a)(x+b) = x^2 + (a+b)x + ab$ (1); $= x^2 + 13x + 40$ (1).	1+1
14	Use the square identity (1); $= 4x^2 - 28xy + 49y^2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Use the square identity (1); $= 4x^2 + 12xy + 9y^2$ (1); LHS $= (2+3)^2 = 25$, RHS $= 25$ (1).	1+1+1
16	Need two numbers with sum 8 and product 12 (1); They are 2 and 6 (1); $= (x+2)(x+6)$ (1).	1+1+1
17	$103 \times 107 = (100+3)(100+7)$ (1); $= 100^2 + (10)(100) + (21)$ (1); $= 11021$ (1).	1+1+1
	OR: Use the square identity (1); $= 9x^2 - 24xy + 16y^2$ (1); LHS $= (3-4)^2 = 1$, RHS $= 1$ (1).	1+1+1
18	Need two numbers with sum -7 and product 10 (1); They are -5 and -2 (1); $= (x-5)(x-2)$ (1).	1+1+1
	OR: $97 \times 95 = (100-3)(100-5)$ (1); $= 100^2 + (-8)(100) + (15)$ (1); $= 9215$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Use $(a-b)^2 = a^2 - 2ab + b^2$ (1); $a = 2x$, $b = 3y$ (1); Squares: $4x^2 + 9y^2$ (1); Cross term: $-12xy$ (1); $= 4x^2 - 12xy + 9y^2$; at $x = 3$, $y = 1$ the value is 9 (1).	1+1+1+1+1
	OR: Use $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ (1); $a = x$, $b = 2$ (1); $3a^2b = 6x^2$, $3ab^2 = 12x$ (1); $= x^3 + 6x^2 + 12x + 8$ (1); Terms checked (1).	1+1+1+1+1
20	Use $(a-b)^2 = a^2 - 2ab + b^2$ (1); $a = 3x$, $b = y$ (1); Squares: $9x^2 + y^2$ (1); Cross term: $-6xy$ (1); $= 9x^2 - 6xy + y^2$; at $x = 2$, $y = 4$ the value is 4 (1).	1+1+1+1+1
	OR: Use $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ (1); $a = 2x$, $b = 1$ (1); $3a^2b = 12x^2$, $3ab^2 = 6x$ (1); $= 8x^3 + 12x^2 + 6x + 1$ (1); Terms checked (1).	1+1+1+1+1

Mistakes to avoid:

- Writing $(a+b)^2 = a^2 + b^2$ and leaving out the middle term $2ab$.
- Using $(a-b)^2 = a^2 - b^2$; this confuses a square of a difference with a difference of squares.
- Sign errors in $(a-b)^3$: the terms alternate in sign, giving $a^3 - 3a^2b + 3ab^2 - b^3$.
- Not squaring the whole term: $(2x+3)^2 = 4x^2 + 12x + 9$, not $2x^2 + 12x + 9$.
- Factorising $x^2 + 5x + 6$ with numbers whose sum or product is wrong; always check that the sum is 5 and the product is 6.
- Not taking out a common factor before applying an identity, e.g. $2x^2 - 18 = 2(x-3)(x+3)$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $x^2 + 3x - 40$	$(x + a)(x + b) = x^2 + (a + b)x + ab$
2	**(d)** 9604	Write $98 = 100 - 2$
3	**(a)** $(x + 6)$	$a^2 + 2ab + b^2 = (a + b)^2$
4	**(c)** $(x - 7)(x - 2)$	Find two numbers with the given sum and product.
5	**(a)** 58	$x^2 + y^2 = (x + y)^2 - 2xy$.
6	**(b)** $(a + b)(a^2 - ab + b^2)$	Sum of cubes identity.
7	**(c)** $25x^2 + 20xy + 4y^2$	$(a + b)^2 = a^2 + 2ab + b^2$
8	**(c)** $16x^2 - 24xy + 9y^2$	$(a - b)^2 = a^2 - 2ab + b^2$
9	**(a)** $x^2 + 7x - 18$	$(x + a)(x + b) = x^2 + (a + b)x + ab$
10	**(c)** 10404	Write $102 = 100 + 2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$96 = 100 - 4$ (1); $96^2 = 9216$ (1).	1+1
12	Use $(x + a)(x + b) = x^2 + (a + b)x + ab$ (1); $= x^2 + 5x - 36$ (1).	1+1
13	Use the square identity (1); $= 36x^2 + 12xy + y^2$ (1).	1+1
14	$105 = 100 + 5$ (1); $105^2 = 11025$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Use the square identity (1); $= 25x^2 + 20xy + 4y^2$ (1); LHS $= (5 + 2)^2 = 49$, RHS $= 49$ (1).	1+1+1
16	Need two numbers with sum 3 and product -10 (1); They are -2 and 5 (1); $= (x - 2)(x + 5)$ (1).	1+1+1
17	$104 \times 98 = (100 + 4)(100 - 2)$ (1); $= 100^2 + (2)(100) + (-8)$ (1); $= 10192$ (1).	1+1+1
	OR: Use the square identity (1); $= 16x^2 - 40xy + 25y^2$ (1); LHS $= (4 - 5)^2 = 1$, RHS $= 1$ (1).	1+1+1
18	Need two numbers with sum -2 and product -24 (1); They are -6 and 4 (1); $= (x - 6)(x + 4)$ (1).	1+1+1
	OR: $106 \times 102 = (100 + 6)(100 + 2)$ (1); $= 100^2 + (8)(100) + (12)$ (1); $= 10812$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Use $(a - b)^2 = a^2 - 2ab + b^2$ (1); $a = 4x$, $b = 5y$ (1); Squares: $16x^2 + 25y^2$ (1); Cross term: $-40xy$ (1); $= 16x^2 - 40xy + 25y^2$; at $x = 2$, $y = 1$ the value is 9 (1).	1+1+1+1+1
	OR: Use $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ (1); $a = x$, $b = 3$ (1); $3a^2b = 9x^2$, $3ab^2 = 27x$ (1); $= x^3 + 9x^2 + 27x + 27$ (1); Terms checked (1).	1+1+1+1+1
20	Use $(a - b)^2 = a^2 - 2ab + b^2$ (1); $a = 5x$, $b = 2y$ (1); Squares: $25x^2 + 4y^2$ (1); Cross term: $-20xy$ (1); $= 25x^2 - 20xy + 4y^2$; at $x = 1$, $y = 3$ the value is 1 (1).	1+1+1+1+1
	OR: Use $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ (1); $a = 3x$, $b = 1$ (1); $3a^2b = 27x^2$, $3ab^2 = 9x$ (1); $= 27x^3 + 27x^2 + 9x + 1$ (1); Terms checked (1).	1+1+1+1+1

Mistakes to avoid:

- Writing $(a + b)^2 = a^2 + b^2$ and leaving out the middle term $2ab$.
- Using $(a - b)^2 = a^2 - b^2$; this confuses a square of a difference with a difference of squares.
- Sign errors in $(a - b)^3$: the terms alternate in sign, giving $a^3 - 3a^2b + 3ab^2 - b^3$.
- Not squaring the whole term: $(2x + 3)^2 = 4x^2 + 12x + 9$, not $2x^2 + 12x + 9$.
- Factorising $x^2 + 5x + 6$ with numbers whose sum or product is wrong; always check that the sum is 5 and the product is 6.
- Not taking out a common factor before applying an identity, e.g. $2x^2 - 18 = 2(x - 3)(x + 3)$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $(x+8)$	$a^2+2ab+b^2=(a+b)^2$
2	**(a)** $(x-3)(x+5)$	Find two numbers with the given sum and product.
3	**(c)** 25	$x^2+y^2=(x+y)^2-2xy$.
4	**(d)** $(a-b)(a^2+ab+b^2)$	Difference of cubes identity.
5	**(d)** $16x^2+8xy+y^2$	$(a+b)^2=a^2+2ab+b^2$
6	**(b)** $25x^2-10xy+y^2$	$(a-b)^2=a^2-2ab+b^2$
7	**(a)** $x^2+2x-24$	$(x+a)(x+b)=x^2+(a+b)x+ab$
8	**(a)** 9409	Write $97=100-3$
9	**(d)** $(x+10)$	$a^2+2ab+b^2=(a+b)^2$
10	**(c)** $(x-4)(x+3)$	Find two numbers with the given sum and product.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Use $(x+a)(x+b)=x^2+(a+b)x+ab$ (1); $=x^2+5x-14$ (1).	1+1
12	Use the square identity (1); $=16x^2-24xy+9y^2$ (1).	1+1
13	$99=100-1$ (1); $99^2=9801$ (1).	1+1
14	Use $(x+a)(x+b)=x^2+(a+b)x+ab$ (1); $=x^2-11x+30$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Use the square identity (1); $=9x^2+6xy+y^2$ (1); LHS $=(3+1)^2=16$, RHS $=16$ (1).	1+1+1
16	Need two numbers with sum -10 and product 21 (1); They are -7 and -3 (1); $=(x-7)(x-3)$ (1).	1+1+1
17	$96\times108=(100-4)(100+8)$ (1); $=100^2+(4)(100)+(-32)$ (1); $=10368$ (1).	1+1+1
	OR: Use the square identity (1); $=36x^2-60xy+25y^2$ (1); LHS $=(6-5)^2=1$, RHS $=1$ (1).	1+1+1
18	Need two numbers with sum 4 and product -21 (1); They are -3 and 7 (1); $=(x-3)(x+7)$ (1).	1+1+1
	OR: $95\times105=(100-5)(100+5)$ (1); $=100^2+(0)(100)+(-25)$ (1); $=9975$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Use $(a-b)^2=a^2-2ab+b^2$ (1); $a=3x$, $b=4y$ (1); Squares: $9x^2+16y^2$ (1); Cross term: $-24xy$ (1); $=9x^2-24xy+16y^2$; at $x=4$, $y=2$ the value is 16 (1).	1+1+1+1+1
	OR: Use $(a+b)^3=a^3+3a^2b+3ab^2+b^3$ (1); $a=2x$, $b=3$ (1); $3a^2b=36x^2$, $3ab^2=54x$ (1); $=8x^3+36x^2+54x+27$ (1); Terms checked (1).	1+1+1+1+1
20	Use $(a-b)^2=a^2-2ab+b^2$ (1); $a=2x$, $b=5y$ (1); Squares: $4x^2+25y^2$ (1); Cross term: $-20xy$ (1); $=4x^2-20xy+25y^2$; at $x=5$, $y=1$ the value is 25 (1).	1+1+1+1+1
	OR: Use $(a+b)^3=a^3+3a^2b+3ab^2+b^3$ (1); $a=x$, $b=4$ (1); $3a^2b=12x^2$, $3ab^2=48x$ (1); $=x^3+12x^2+48x+64$ (1); Terms checked (1).	1+1+1+1+1

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